

Final Exam

3 June 2020

Exam Instructions

- The exam is an „open-book online exam “, which means that you are allowed to use your lecture and exercise materials as well as a calculator when solving the exam problems.
- With the download of the exam, you commit yourself to solving your exam problems on your own. It is not allowed to solve the exam problems in groups.
- The exam material consists of the following two files that you have to download at the beginning of the exam time:
 1. One PDF file (.pdf) that includes the problem set of the exam (including exam instructions). This PDF file consists of 8 pages.
 2. One solution file in Excel format (.xlsx), in which you have to fill in your solutions (see also the instructions in section „Solution File Instructions“).
- **Important: At the end of the exam time, you are allowed to upload only the solution file. When correcting the exams, we only consider solutions that were filled into the solution file in the required way.**
- There are a large number of different exam versions. The different versions are marked with numbers. The number at the bottom left identifies your exam version.
- The exam consists of 30 questions. On the problem set, you find 5 answer possibilities for each exam question. For each question, **exactly one** of the given answer possibilities is correct. Here, “correct” means that the statement is true in each case that is covered by the underlying model that was part of the course curriculum.
- You have **120 Minuten** to solve the exam.
- The time window for solving the exam starts at 8:30am and ends at 10:30am. You have 10 additional minutes for downloading the exam and 10 additional minutes to upload the solution file. This means that you can download the exam from 8:20am, and the uploading of your solution file **has to be completed by 10:40am**.
- If the download of the exam materials is not possible because of technical problems (e.g. in case that you are not able to enter the exam website), then you must send an e-mail to the address **mikroa@mail.uni-mannheim.de**, indicating your full name and your student ID (Matrikelnummer). You must send this email from your official university-e-mail account; we will not consider e-mails sent from private e-mail addresses. In your e-mail, you must precisely explain the technical problem that occurred. Then, we will try to send you the exam materials as soon as possible by e-mail. **Important: please send your e-mail only after you have unsuccessfully tried to download the exam materials several times.**
- If the upload of your solution file is not possible because of technical problems (e.g. in case that you are not able to enter the exam website), then you must send the solution file to the e-mail address **mikroa@mail.uni-mannheim.de**, indicating your full name and your student ID (Matrikelnummer). You must send this email from your official university-e-mail account; we will not consider e-mails sent from private e-mail addresses. In your e-mail, you must precisely explain the technical problem that occurred. A solution file sent by e-mail due to a technical problem can be considered only if we receive it until 10:40am the latest. Any solution file that we receive after 10:40am will get assigned the grade 5.0.

- In case that, at the end of the exam time, your computer has no internet access anymore, you should try to upload your solutions using a mobile device (or, if this is not possible, by e-mail, see above). In case that you find no device to upload the exam, you must document the technical problem accurately (e.g. by using screen shots or by taking a photo etc.) and contact us immediately by telephone using the following number: 0621-181-3428. Important: We only document the problem, but the decision how to proceed with your exam will be subject to the examination board (Prüfungsausschuss) (for this reason, it is important to document the problem accurately). **You are not allowed to use this phone number to ask questions concerning the content of the exam or any other technical questions.**

Solution File Instructions

- The **solution file** has an Excel format (.xlsx).
- Save the Excel file directly on your computer after you have downloaded it. Please also save the file regularly after you have entered solutions in order to avoid any file loss during the exam time.
- In the second line (in yellow) you have to fill in your personal information as well as the respective solutions.
- You fill in the following information: in **cell A2 your student ID (Matrikelnummer)**, in **cell B2 your user name** (user name of your university account; i.e. if your university e-mail address is e.g. *mmustermann@mail.uni-mannheim.de*, then you have to enter *mmustermann*) and in **cell C2 the version of your exam** (if your exam indicates e.g. *Version 10* at the bottom, then you have to fill in **10** here).
- You have to fill in the solutions of the questions in the corresponding cells, i.e. you have to fill in the solution of question 1 (Q1) in cell D2, the solution of question 2 (Q2) in cell E2 etc.
- Only the following answer possibilities are valid: {**a, b, c, d, e**} (Important: use only lower case letters as indicated!). Only answers of this form will be graded.
- Important: You are only allowed to use the cells in the second line (are in yellow) and you are not allowed to delete or add lines or columns! If you change the structure of the Excel file by deleting or adding lines or columns, then we are not able to grade your exam.
- After you have filled in your solutions, save the solution file using the existing Excel format (.xlsx) and your student ID (Matrikelnummer) as file name, i.e. the name of the solution file should be "[YourStudentID].xlsx" (d.h. if your student ID is *123456*, then you save the file as *123456.xlsx*). Use this file name when uploading the solution file.)

Grading

- In total you can obtain up to *30 points*.
- For each correct answer, you obtain *one point*.
- If you obtain at least *27 points*, you get the grade 1.0 in any case.
- You will pass the exam with certainty if you obtain at least *14 Punkte* or if you are among the 75% best participants of the exam.

Good Luck!

1. Consider an Edgeworth exchange economy. One half of the consumers has preferences that are represented by the utility function $u^A(x_1, x_2) = (x_1)^c x_2$, where $c > 0$ is a given parameter. The other half of the consumers has preferences that are represented by the utility function $u^B(x_1, x_2) = \min\{x_1, x_2\}$. Every consumer has an initial endowment of $(10, 10)$. The prices of the goods are denoted p_1 and p_2 . After plugging in all individual demands, the market clearing condition for good 1 can be written as follows:

- (a) $\frac{c}{c+1} \left(1 + \frac{p_2}{p_1}\right) + 1 = 2.$
- (b) $\frac{1}{c+1} \left(1 + \frac{p_2}{p_1}\right) + 1 = 2.$
- (c) $\frac{1}{c+1} \left(1 + \frac{p_2}{p_1}\right) + 10 = 20.$
- (d) $\frac{c}{c+1} \left(1 + \frac{p_2}{p_1}\right) + 10 = 20.$
- (e) $\frac{c}{c+1} \left(1 + \frac{p_1}{p_2}\right) + 10 = 20.$

2. A consumer has Cobb-Douglas preferences. The marginal rate of substitutions at the point $(x_1, x_2) = (3, 2)$ equals $-4/5$. The preferences can be represented by the following utility function:

- (a) $u(x_1, x_2) = (x_1)^{12}(x_2)^{10}$
- (b) $u(x_1, x_2) = (x_1)^{10}(x_2)^{12}$
- (c) $u(x_1, x_2) = (x_1)^8(x_2)^{15}$
- (d) $u(x_1, x_2) = (x_1)^{15}(x_2)^8$
- (e) None of the other answers is correct.

3. A consumer's preferences are represented by the utility function $u(x_1, x_2) = \max\{x_1, 3x_2\}$. If the price of good 2 is three times the price of good 1, that is, $3p_1 = p_2$, and if Y denotes the consumer's income, then the set of utility-maximizing bundles equals

- (a) $\{(0, \frac{Y}{3p_1}), (\frac{Y}{p_1}, 0)\}$
- (b) $\{\lambda(0, \frac{Y}{3p_1}) + (1 - \lambda)(\frac{Y}{p_1}, 0) \mid 0 \leq \lambda \leq 1\}$
- (c) $\{(\frac{Y}{3p_2}, 0), (0, \frac{Y}{p_2})\}$
- (d) $\{\lambda(0, \frac{Y}{p_1}) + (1 - \lambda)(\frac{Y}{3p_1}, 0) \mid 0 \leq \lambda \leq 1\}$
- (e) $\{\lambda(0, \frac{Y}{p_2}) + (1 - \lambda)(\frac{Y}{3p_2}, 0) \mid 0 \leq \lambda \leq 1\}$

4. Jevon's law is violated in a market if

- (a) there are many traders in this market
- (b) all market transactions take place on the same market platform
- (c) the price or prices used in the various transactions are fully transparent
- (d) the market clears
- (e) the price used in one half of the market transactions is considerably higher than the price used in the other half of the transactions

5. Consider an exchange economy with two goods and 30 traders. 20 traders have preferences that are represented by the utility function $u^A(x_1, x_2) = x_1 x_2$. Each of these traders has the initial endowment $(1, 0)$. The remaining 10 traders have preferences that are represented by the utility function $u^B(x_1, x_2) = x_1(x_2)^d$, where $d > 0$ is a given parameter. Each of these traders has the initial endowment $(0, 2)$. There exists a competitive equilibrium in which the price of both goods is the same if $d =$

- (a) 1
- (b) 7
- (c) 3
- (d) 9
- (e) 11

6. Consider an exchange economy with two goods and two individuals, Amelie and Ben. Each individual has the initial endowment $(30, 70)$. Amelie's preferences are represented by the utility function $u^A(x_1, x_2) = x_1 + x_2$. Ben's preferences are represented by the utility function $u^B(x_1, x_2) = (x_1)^c x_2$, where $c > 0$ is a given parameter. The allocation in which both traders keep their initial endowments is a Pareto efficient allocation if $c =$
- $3/7$
 - $3/4$
 - $4/3$
 - $7/4$
 - $4/7$
7. A firm with the following production function has a non-convex technology:
- $f(x_1, x_2) = \min\{x_1 + x_2, 2x_2\}$
 - $f(x_1, x_2) = \max\{x_1, x_2\}$
 - $f(x_1, x_2) = \min\{x_1, x_2\}$
 - $f(x_1, x_2) = x_1 + x_2/7$
 - $f(x_1, x_2) = x_1 + \sqrt{x_2}$
8. Let $f(x_1)$ be a strictly increasing production function with the single input x_1 . Suppose that $f(0) = 0$, $MP_1(0) = 0$, the marginal product of the input is strictly increasing in the area from $x_1 = 0$ to $x_1 = 7$, and starting from the point $x_1 = 7$ the marginal product is strictly decreasing with $\lim_{x_1 \rightarrow \infty} MP_1(x_1) = 0$. Let x^D denote the point where the average product is maximal. Then the following holds:
- $0 < x^D < 7$
 - $x^D > 7$
 - $x^D = 0$
 - $x^D = 7$
 - x^D can be any non-negative number
9. A firm has the production function $f(x_1, x_2) = x_1 + (x_2)^2$. The input prices are $p_1 = p_2 = 1$. Let $C(q)$ denote the firm's cost function. Then the following holds.
- $C(q) = \sqrt{q}$ for all $q > 1$
 - $C(q) = \sqrt{q}$ for all $q < 1$
 - $C(q) = q^2$ for all $q > 1$
 - $C(q) = q^2$ for all $q < 1$
 - None of the other answers is correct.
10. A firm has setup costs of 10. In the area $0 < q < 10$, the marginal cost function satisfies the inequality $1 < MC(q) < 2$. Then the following holds for the average cost function at the point $q = 10$:
- $2 < AC(10) < 3$
 - $0 < AC(10) < 1$
 - $1 < AC(10) < 2$
 - $3 < AC(10) < 4$
 - none of the other answers is correct
11. If firm's production function has decreasing returns to scale, then the average-cost function is
- strictly increasing
 - strictly decreasing
 - U-shaped
 - constant
 - none of the other answers is correct

12. A firm has the average-cost function $AC(q) = 2 + (q - 1)^2$. Then the following holds:
- (a) At any output price larger than 2, the profit-maximizing quantity is larger than 1.
 - (b) At any output price larger than 1, the profit-maximizing quantity is larger than 2.
 - (c) At any output price larger than $1/2$, the profit-maximizing quantity is larger than 1.
 - (d) At any output price larger than $1/2$, the profit-maximizing quantity is larger than 2.
 - (e) none of the other answers is correct
13. A firm has the marginal-cost function $MC(q) = q^3$ and has the setup cost 5. Then the firm's cost function is
- (a) $C(q) = q^4/4 + 5$.
 - (b) $C(q) = q^4/4$
 - (c) $C(q) = q^2$
 - (d) $C(q) = (q - 5)^3 + 5$
 - (e) $C(q) = q^2 + 5$
14. Consider a market supply function $S(p)$ such that $S(20) = 80$ and the price elasticity $\eta(20) = 1$. Using this elasticity, the quantity supplied at the price 21 can be estimated to be approximately equal to
- (a) 84
 - (b) 82
 - (c) 78
 - (d) 76
 - (e) 80
15. In a quasilinear economy there are 20 consumers. Each consumer $i = 1, \dots, 20$ has the willingness-to-pay function $v^i(x^i) = (x^i)^{1/3}$, where x^i is the number of units consumed of good 1. The market demand for good 1 is
- (a) $D(p) = 20(1/(3p))^{3/2}$
 - (b) $D(p) = 20(3p)^{3/2}$
 - (c) $D(p) = 20(1/(3p))^{2/3}$
 - (d) $D(p) = 20(3p)^{2/3}$
 - (e) none of the other answers is correct
16. Consider a quasilinear economy with free entry. Every firm has the average-cost function $AC(q) = 5 + (q - 3)^4$. Then the market supply quantity at the price $p = 5$ is an integer multiple of the number
- (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
 - (e) 5
17. Suppose that, in a quasilinear economy, the demand increases and the supply decreases. Then the following *cannot* happen:
- (a) The equilibrium price decreases.
 - (b) The equilibrium quantity decreases.
 - (c) The equilibrium price increases.
 - (d) The equilibrium quantity increases.
 - (e) None of the other answers is correct.

18. Consider a quasilinear economy in which the market supply is $S(p)$ and the market demand is $D(p)$. Suppose that a subsidy of 5 per unit of good traded is paid to the respective buyer. Then the following equation holds for the equilibrium consumer price p^D .
- (a) $D(p^D) = S(p^D + 5)$
 - (b) $D(p^D) = S(p^D - 5)$
 - (c) $D(p^D + 5) = S(p^D)$
 - (d) $D(p^D - 5) = S(p^D)$
 - (e) None of the other answers is correct.
19. Consider a quasilinear economy with the market supply $S(p) = \max\{0, 3p - 2\}$ and the market demand $D(p) = \max\{0, 5 - 4p\}$. We denote by t the smallest tax rate such that the equilibrium quantity equals 0. Then the following holds:
- (a) $t = 7/12$
 - (b) $t = 11/12$
 - (c) $t = 1$
 - (d) $t = 5/12$
 - (e) $t = 3/12$.
20. Consider a quasilinear economy with the market supply $S(p) = \max\{0, 5p - 4\}$ and the market demand $D(p) = \max\{0, 3 - 2p\}$. In equilibrium the aggregate welfare is equal to
- (a) $7/20$
 - (b) $1/3$
 - (c) $1/24$
 - (d) $9/40$
 - (e) $1/5$.
21. Consider a quasilinear economy with the market supply $S(p) = \max\{0, 5p - 4\}$ and the market demand $D(p) = \max\{0, 4 - 3p\}$. In equilibrium the aggregate consumer surplus is equal to
- (a) $1/6$
 - (b) $1/10$
 - (c) $1/4$
 - (d) $1/3$
 - (e) $1/2$.
22. A consumer's preferences over consumption streams (c_0, c_1) are represented by the utility function $u(c_0, c_1) = \min\{c_0, \frac{1}{11}c_1\}$. The consumer's initial endowment is the income stream $(Y_0, Y_1) = (1, 0)$. The consumer can save money at the interest rate $r = 10\%$. The following consumption quantity is optimal in period 1:
- (a) 1
 - (b) 11
 - (c) $1/10$
 - (d) $1/11$
 - (e) 0.
23. A consumer's preferences over consumption streams (c_0, c_1) are represented by the utility function $u(c_0, c_1) = \sqrt{c_0} + \frac{1}{2}\sqrt{c_1}$. The consumer can save and borrow money at the interest rate r . At which interest rate r is the consumer indifferent between obtaining the initial endowment $(2, 7)$ and receiving the initial endowment $(3, 1)$?
- (a) $r = 300\%$
 - (b) $r = 400\%$
 - (c) $r = 500\%$
 - (d) $r = 600\%$
 - (e) $r = 700\%$.

24. A consumer's preferences over consumption bundles (x_1, x_2) are represented by the utility function $u(x_1, x_2) = \min\{3x_1, 5x_2\}$. The price of good 1 is equal to 6. The consumer's income is equal to 1. Then the consumer's demand function for good 2 is given as follows:

- (a) $p_2 \mapsto \frac{1}{10+p_2}$
- (b) $p_2 \mapsto \frac{1}{14+p_2}$
- (c) $p_2 \mapsto \frac{1}{5+p_2}$
- (d) $p_2 \mapsto \frac{1}{3p_2}$
- (e) $p_2 \mapsto \frac{1}{p_2}$.

25. Referring to the good the price of which is changed, the substitution effect of a price decrease is always

- (a) ≤ 0 .
- (b) < 0 .
- (c) ≥ 0 .
- (d) > 0 .
- (e) $= 0$.

26. A consumer's preferences over consumption bundles is represented by the utility function $u(x_1, x_2) = (x_1)^4(x_2)^5$. The consumer's income is given by the market value of her initial endowment $(6, 7)$. Let p_1 and p_2 denote the prices of the goods. The consumer will buy additional units of good 1 (i.e., she will consume more than 6 units of good 1) if and only if

- (a) $14p_2 > 15p_1$.
- (b) $35p_2 > 24p_1$.
- (c) $10p_2 > 21p_1$.
- (d) $6p_2 < p_1$.
- (e) $6p_2 = p_1$.

27. Suppose that, in the leisure-consumption model, k denotes the consumption spending of a consumer, l denotes her leisure time, T her time budget, k_0 her available income if she does not work, and w her wage per unit of work time. Then her budget constraint is given by the following combination of inequalities:

- (a) $k + wl \leq k_0 + wT$ und $l \leq T$
- (b) $k + wl \leq k_0 + T$ und $wl \leq T$
- (c) $k + wl \leq k_0 + wT$ oder $l \leq T$
- (d) $k + wl \leq k_0 + T$ oder $wl \leq T$
- (e) $k + wl \leq k_0 + wT + T$ und $wl + l \leq T$

28. A consumer has perfect-substitute preferences over bundles (x_1, x_2) (with $x_i \geq 0$ for $i = 1, 2$). The following utility function does *not* represent the consumer's preferences:

- (a) $u(x_1, x_2) = (x_1)^2 + (x_2)^2$
- (b) $u(x_1, x_2) = (x_1 + x_2)^2$
- (c) $u(x_1, x_2) = \sqrt{x_1 + x_2}$
- (d) $u(x_1, x_2) = \min\{x_1, x_2\} + \max\{x_1, x_2\}$
- (e) $u(x_1, x_2) = x_1 + x_2$

29. A consumer has monotonic and convex preferences $\succeq$ over consumption bundles (x_1, x_2) (with $x_i \geq 0$ for $i = 1, 2$). Assume that $(3, 6) \succeq (7, 3)$ and $(7, 3) \succeq (3, 6)$. Then the following holds:

- (a) $(7, 3) \succeq (8, 2)$
- (b) $(4, 4) \succeq (7, 3)$
- (c) $(3, 5) \succeq (8, 2)$
- (d) $(4, 4) \succeq (8, 2)$
- (e) $(3, 5) \succeq (7, 3)$

30. A consumer has the utility function $u(x_1, x_2) = (x_1)^c (x_2)^d$ over bundles (x_1, x_2) , where $c > 0$ and $d > 0$ are given parameters. At each point (x_1, x_2) with $x_1 > 0$ and $x_2 > 0$, the consumer has the following marginal rate of substitutions of good 2 for good 1:

- (a) $MRS_{12}(x_1, x_2) = -\frac{c}{d} \frac{x_2}{x_1}$
- (b) $MRS_{12}(x_1, x_2) = -\frac{c}{d} \frac{x_1}{x_2}$
- (c) $MRS_{12}(x_1, x_2) = \frac{c}{d} \frac{x_2}{x_1}$
- (d) $MRS_{12}(x_1, x_2) = \frac{c}{d} \frac{x_1}{x_2}$
- (e) $MRS_{12}(x_1, x_2) = -\frac{c}{c+d}$